Rule-Based Monetary Policy in China

A Counterfactual Analysis Using the Wolf et al. (2025) Framework

Liming Lin

April 10, 2026

Sciences Po Paris

Supervisor: Paul Bouscasse

Motivation

The puzzle. Since 2023, Chinese CPI inflation has hovered near zero, persistently undershooting the official targets set in the annual Government Work Report (GWR).

- GWR CPI targets 2023–2025: **3%, 3%, 2%**
- Realized CPI YoY 2023–2025: roughly **0%**
- Xi's speech at the 2024 Central Economic Work Conference first introduced the phrase "a reasonable rebound in the price level" (物价合理回升); the 2025 CEWC explicitly tied this objective to the conduct of monetary policy.

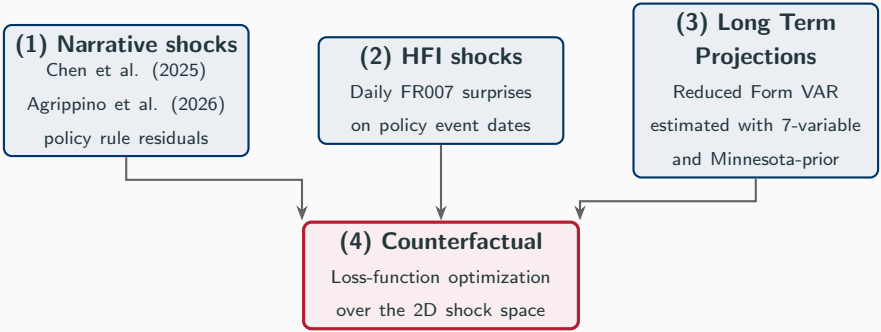
The question. What would have happened to the Chinese economy if the PBoC had strictly followed an inflation-targeting rule during this period?

Why it matters.

- Quantifies the policy gap: how far was actual policy from target-consistent policy?
- Tests whether conventional interest rate policy is even capable of closing the inflation gap, given China's transmission mechanism.

Methodology Roadmap

Following Caravello, McKay & Wolf (2025), the empirics-only counterfactual requires **four building blocks**:



Methodology Roadmap Explanation

- Blocks (1) and (2) provide **two distinct empirical IRFs** of the Chinese economy to monetary policy shocks — the policy "treatments" whose linear combinations span the feasible policy space.
- Block (3) provides the **baseline projection** of where the economy goes absent any policy intervention.
- Block (4) solves for the shock sequence that moves the baseline toward an inflation-targeting path, subject to a multi-objective loss function.

Narrative Shocks: Estimated Policy Rule

Specification. A monthly policy rule for the 7-day interbank repo rate (FR007), combining elements from Chen, Xiao & Zha (2025) and Miranda-Agrippino, Nenova & Rey (2026):

$$\begin{aligned}\text{FR007}_t \;=\; \alpha \;+\; \rho \cdot \text{FR007}_{t-1} \;+\; \beta_\pi \cdot \text{cpi_gap}_{t-1} \\ +\; \beta_y^+ \cdot \text{gap_pos}_{t-1} \;+\; \beta_y^- \cdot \text{gap_neg}_{t-1} \;+\; \beta_e \cdot \text{fx_gap}_{t-1} \;+\; \varepsilon_t^{\text{MP}}\end{aligned}$$

Variable definitions.

- $\text{cpi_gap}_t = \pi_t - \pi_t^*$, where π_t^* is the annual CPI target from the Government Work Report
- $\text{gdp_gap}_t = g_t - g_t^*$, where g_t is monthly real GDP YoY (see appendix: ▸ Stock–Watson construction) and g_t^* is the GWR growth target
- gap_pos_t, gap_neg_t: asymmetric split of the GDP gap around a 1pp threshold (see appendix: ▸ regime-switch motivation)
- fx_gap_t: post-2006 interaction between the change in the central parity rate and the spot–parity deviation

Narrative Shocks: Identification and Rationale

Identification. The residual $\widehat{\varepsilon}_t^{\mathrm{MP}}$ isolates the unsystematic component of FR007 movements — the narrative monetary policy shock series. ▸ actual vs. predicted FR007

Why this hybrid specification?

- CRZ/Chen kept FR007 as the dependent variable but excluded the exchange rate; MANR2026 added the FX interaction but used the CMPI composite on the left-hand side.
- We retain FR007 (the actual market rate that anchors HFI surprises downstream) while adopting MANR2026's FX specification.

High-Frequency Shocks: FR007 Surprises on Policy Event Dates

Identification strategy. Following Das & Song (2022), we identify monetary policy surprises as the daily close-to-close change in the FR007 rate around dates on which the PBoC announces a change to one of its core policy instruments:

$$\text{shock}_t \;=\; \left(\text{FR007}_t - \text{FR007}_{t-1}\right) \cdot \mathbb{1}\{t \in \mathcal{E}\}$$

where $\mathcal{E}$ is the set of policy event dates.

Core policy instruments tracked. Event dates are hand-collected from PBoC announcements (compiled in CMPI.xlsx, Miranda-Agrippino, Nenova & Rey 2026):

- 7-day reverse repo rate (OMO) — the primary short-term policy rate post-2016
- 1-year Medium-term Lending Facility (MLF) — introduced 2014, principal medium-term policy tool
- 1-year Loan Prime Rate (LPR) — replaced benchmark lending rate as the loan pricing anchor in August 2019
- Reserve Requirement Ratio for large financial institutions (RRR)

Reduced-Form Long-Term Projection: Bayesian VAR

extbfSpecification. A 7-variable monthly BVAR with $p = 6$ lags, estimated with a Minnesota-type prior following Giannone, Lenza & Primiceri (2015):

$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + \sum_{k=1}^{4} d_k \, D_t^{\text{covid},k} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u)$$

where $D_t^{\text{covid},k}$ are monthly dummies for Jan–Apr 2020.

extbfVariables in y_t (BVAR ordering, monthly).

$$y_t = \left(g_t,\ \text{IP}_t,\ \pi_t,\ i_t,\ m_t,\ e_t,\ y_t^{US}\right)'$$

g_t: real GDP YoY (Stock–Watson distribution from quarterly NBS GDP, appendix); IP_t: industrial production YoY; π_t: CPI YoY; i_t: FR007; m_t: M2 YoY growth; e_t: RMB NEER YoY; y_t^{US}: US industrial production YoY.

Reduced-Form Long-Term Projection: Baseline and MA Representation

extbfEstimation samples.

- Monthly estimation with COVID dummies included in all runs.
- For each projection exercise, parameters are estimated through $t^* - 1$.
- Baseline vintages reported for 2019, 2022, and 2025.

extbfBaseline projection.

$$\hat{y}_{t^*+h} = \hat{c} + \sum_{\ell=1}^{6} \hat{A}_\ell \, \hat{y}_{t^*+h-\ell}, \qquad h = 1, 2, \ldots, H$$

extbfNo data from the projection window enters the forecast. For $\ell < h$, $\hat{y}_{t^*+h-\ell}$ is previously forecast; for $\ell \geq h$, it is pre-t^* realized data.

extbfStored for counterfactual stage: Wold MA form $y_t = c^* + \sum_{h=0}^{120} \Psi_h \, u_{t-h}$.

Appendix — Asymmetric GDP Gap Specification

Construction. Define the threshold indicator $\mathbb{1}_t^- = \mathbb{1}\{\text{gdp_gap}_t < 1\}$, then split:

$$\text{gap_pos}_t = \text{gdp_gap}_t \cdot (1 - \mathbb{1}_t^-), \qquad \text{gap_neg}_t = \text{gdp_gap}_t \cdot \mathbb{1}_t^- .$$

Motivation. Following Chen, Ren & Zha (2018), the PBoC's response to growth is hypothesised to be *asymmetric*: monetary policy reacts more aggressively when realized GDP growth falls meaningfully below the official Government Work Report target than when it merely fluctuates around it.

Why a 1pp threshold rather than zero?

- Chinese GWR growth targets are widely understood as floors rather than point estimates — modest overshoots are routine and politically uncontroversial.
- A small undershoot ($0 \leq \text{gap} < 1$) is treated as a near-miss and rarely triggers a policy response.
- Only when the shortfall exceeds roughly one percentage point does the PBoC enter a clearly accommodative regime.

Appendix — Policy Rule Fit and Residuals

Top. Actual FR007 vs. predicted FR007 from the rule regression. **Bottom.** Estimated residual series $\widehat{\varepsilon}_t^{\mathrm{MP}}$, used as the narrative monetary policy shock.

Robustness: regression re-estimated excluding 2020 to verify that COVID-period observations do not unduly drive the coefficient estimates.

back

Appendix — Monthly GDP Construction

[To be filled in: Stock & Watson (2010) distribution method, adapted to NBS quarterly GDP (当季值) using the sum identity $Q_T = q_{3T-2} + q_{3T-1} + q_{3T}$, with monthly indicators consumption / FAI / government spending / trade balance.]

back

Appendix — HFI Shocks: IRF Comparison

Left. IRF of Chinese economy to monetary policy shocks identified from FR007 surprises on *any* PBoC policy announcement day. **Right.** IRF using *rate-change-only* identification — cleaner shock, smaller sample size.

The rate-change-only series serves as the primary instrument in the counterfactual analysis, with the any-announcement series used for robustness checks.

back